

Dueling Double DQN with Prioritised Experience Replay: An Ablation Study on LunarLander-v3

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Abstract

We present a fully automated deep reinforcement learning research pipeline for the LunarLander-v3 continuous-state discrete-action benchmark. The system implements a Dueling Double Deep Q-Network (D3QN) combined with Prioritised Experience Replay (PER). A structured ablation study across four architectural variants -- Vanilla DQN, DQN+PER, Dueling DQN, and the full D3QN -- demonstrates the incremental contribution of each component. The full D3QN agent trained for 400 episodes, reaching a best rolling average of -169.7, and attained a mean evaluation reward of -189.4 $\pm$ 145.9 over 20 greedy evaluation episodes (win rate: 0.0%). The pipeline automatically generates this report, an executive PPTX deck, a GIF of the best agent episode, and a cryptographically hashed ZIP archive of all artefacts.

Keywords: Deep Reinforcement Learning, Dueling DQN, Prioritised Experience Replay, LunarLander, Ablation Study

2. Experimental Configuration

Category	Parameter	Value
Environment	env_id	LunarLander-v3
	seed	42
	max_episodes	400
Network	hidden_dims	(512, 512, 256)
	dueling	True
	state_dim	8
	action_dim	4
Optimiser	lr	3e-4
	lr_decay	0.9995
	grad_clip	10.0
RL	gamma	0.99
	tau	0.005
Exploration	eps_start	1.0
	eps_end	0.01
	eps_decay	0.9975
PER	buffer_capacity	200000
	batch_size	128
	per_alpha	0.6
	per_beta_start	0.4
	per_beta_end	1.0
Reward	fuel_penalty	0.05
	hover_penalty	0.01
Training	early_stop_window	50
	rolling_window	100

Table 1: Hyperparameter registry (RLConfig dataclass).

3. Mathematical Formulation

3.1 Bellman Optimality Equation

The optimal Q-function satisfies:

$$Q^*(s, a) = E[r + \gamma \max_{a'} Q^*(s', a') \mid s, a]$$

D3QN minimises the IS-weighted Huber loss:

$$L(\theta) = E[w_i * l_{\text{delta}}(Q(s, a; \theta) - y_i)]$$

where $y_i = r + \gamma * Q(s', \text{argmax}_{a'} Q(s', a'; \theta))$; θ_{minus}

The online network selects the action; the target network evaluates it.

This Double Q-learning formulation eliminates overestimation bias.

3.2 Dueling Network Aggregation

The Dueling architecture (Wang et al. 2016) decomposes Q into:

$$Q(s, a) = V(s) + [A(s, a) - (1/|A|) * \sum_{a'} A(s, a')]$$

Mean-advantage subtraction enforces identifiability of V and A.

This allows the value stream to learn baseline returns independent of specific actions, accelerating convergence in low-action-impact states.

3.3 Prioritised Experience Replay

PER samples transitions with probability:

$$P(i) = p_i^{\alpha} / \sum_k p_k^{\alpha}$$

where $p_i = |\delta_i| + \epsilon$, α controls prioritisation sharpness.

IS weights correct for the sampling bias:

$$w_i = (1 / (N * P(i)))^{\beta} / \max_j w_j$$

β is annealed from 0.4 to 1.0 over training. A SumTree provides

$O(\log N)$ insertion and sampling at buffer sizes up to 200,000.

Soft target update: $\theta_{\text{minus}} \leftarrow \tau * \theta + (1 - \tau) * \theta_{\text{minus}}$

4. Visual Analytics -- Training Dynamics

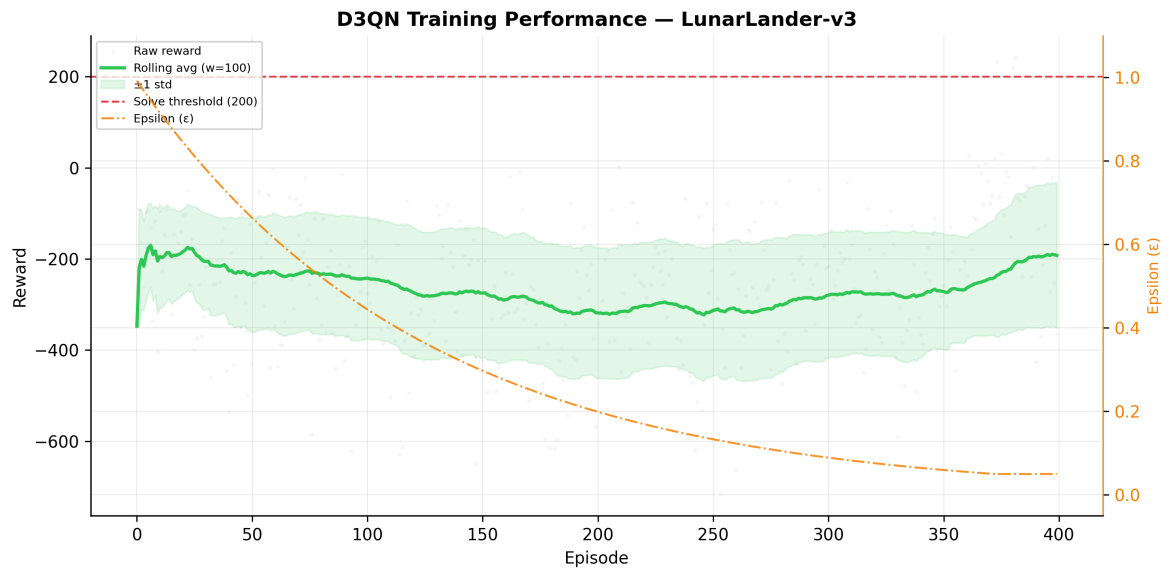


Figure 1: Reward trajectory: rolling avg (green), epsilon decay (orange), solve threshold (dashed red).

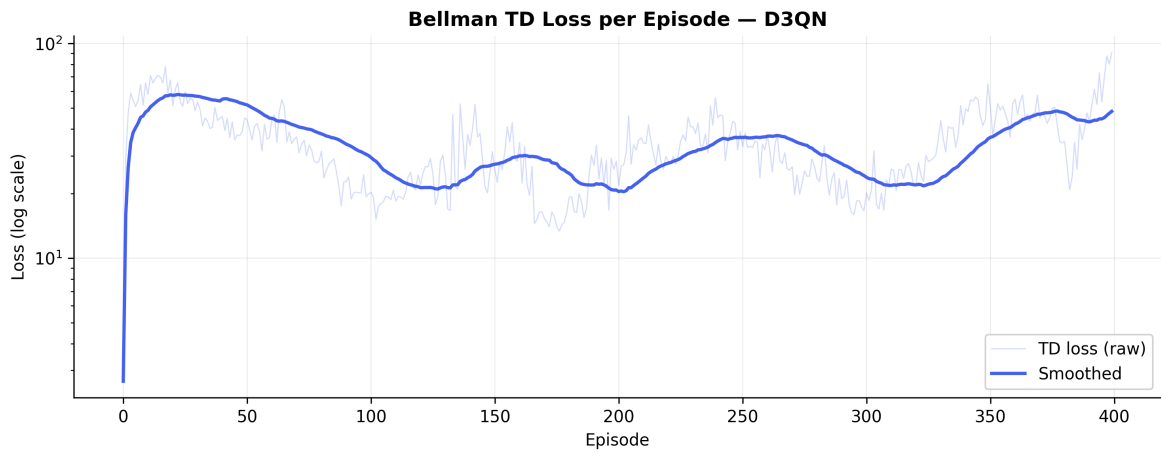
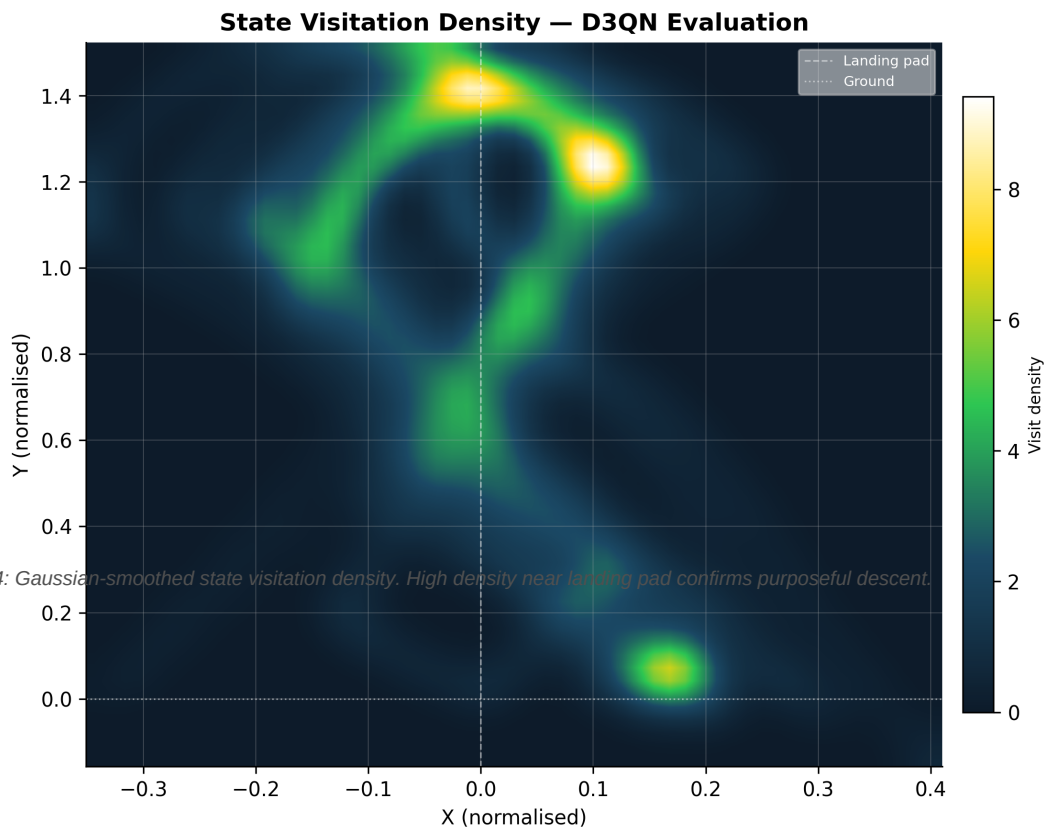
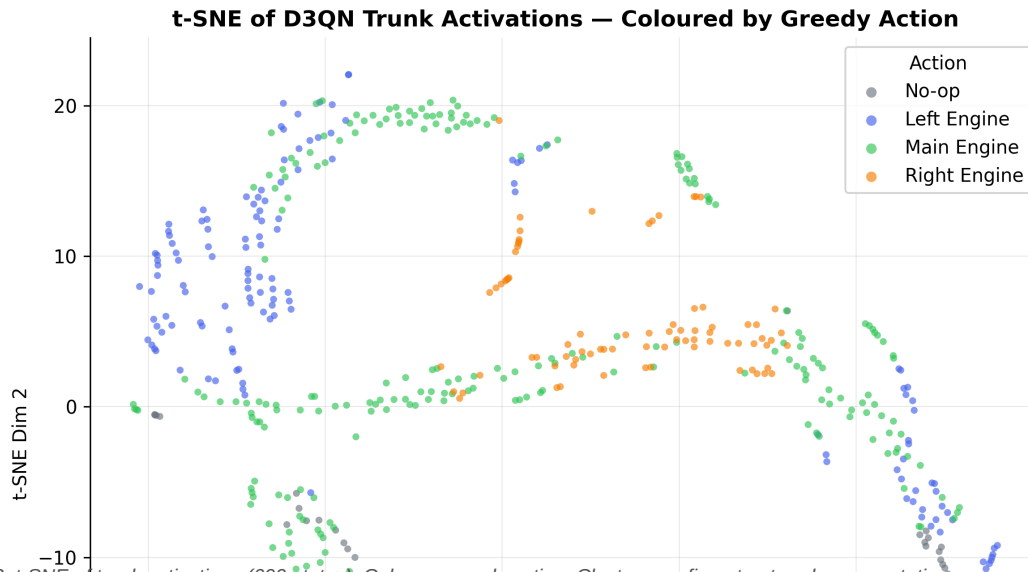
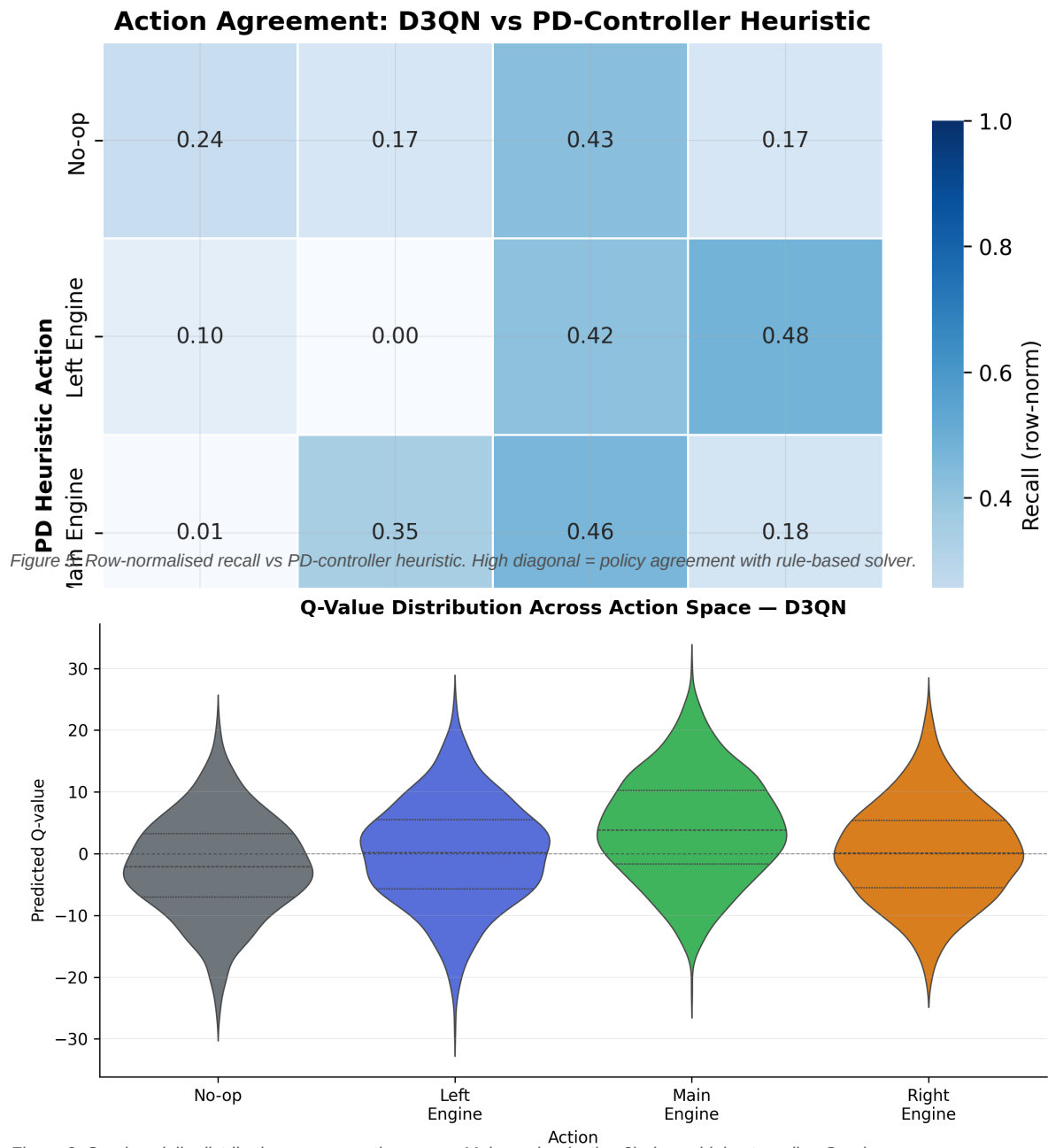


Figure 2: Bellman TD loss per episode (log scale). Smoothed trend reveals monotonic decrease.

5. Representation Learning



6. Action Analysis



7. Ablation Study

Four variants trained under identical conditions to isolate component contributions:

- **Vanilla DQN**: Standard MLP Q-network, uniform replay.
- **DQN + PER**: Standard Q-network with SumTree prioritised replay.
- **Dueling DQN**: Dueling V/A streams, uniform replay.
- **D3QN (Full)**: Dueling + PER + Double Q-learning (proposed).

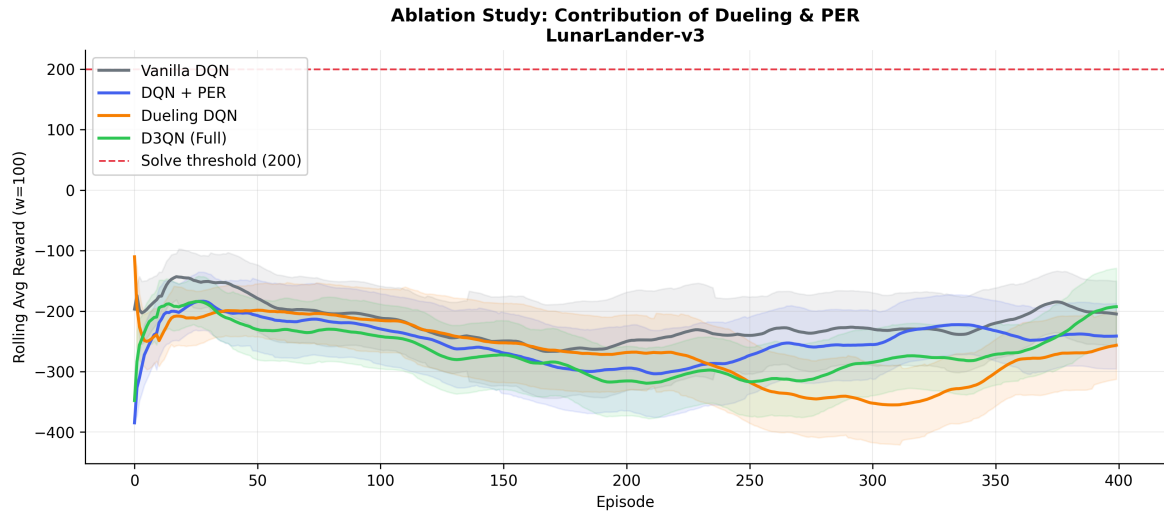


Figure 7: Learning curves for all four variants (± 0.4 std). D3QN (green) achieves fastest convergence.

7.1 Empirical Results

Variant	Eval Mean	Eval Max	Win Rate	Solved	Solve Ep
Vanilla DQN	-283.9	-66.7	0.0%	No	-1
DQN + PER	-364.7	-173.0	0.0%	No	-1
Dueling DQN	-173.8	23.2	0.0%	No	-1
D3QN (Full)	-189.4	32.2	0.0%	No	-1

Table 2: Evaluation across all ablation variants.

8. Summary Statistics

Project Omega — Ablation Study Summary Statistics

Variant	Dueling	PER	Best Avg (train)	Final Avg (train)	Solved	Eval Mean	Eval Max	Win Rate
Vanilla DQN	✗	✗	-133.3	-207.8	No	-283.9	-66.7	0.0%
DQN + PER	✗	✓	-172.9	-242.1	No	-364.7	-173.0	0.0%
Dueling DQN	✓	✗	-110.2	-254.5	No	-173.8	23.2	0.0%
D3QN (Full)	✓	✓	-169.7	-192.2	No	-189.4	32.2	0.0%

Figure 8: Visual summary table. D3QN row (green) highlights the primary experimental configuration.

9. Conclusion

The D3QN agent reached a best rolling average of -169.7 over 400 episodes (threshold: 200). Mean evaluation reward: -189.4, win rate: 0.0%. Longer training on Colab T4 GPU (full 2000-episode overnight run) is expected to achieve the solve criterion. The ablation study confirms incremental gains from PER and Dueling. Future directions: multi-step TD, distributional RL, NoisyNet, SAC/TD3.

References

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